Chapter 98

Effect Nodes

This chapter details the Effect nodes in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Duplicate [Dup]

The Duplicate node

Duplicate Node Introduction

Similar to the Duplicate 3D node, the Duplicate node can be used to quickly duplicate any 2D image, applying a successive transformation to each, and creating repeating patterns and complex arrays of objects. The options in the Jitter tab allow for non-uniform transformations, such as random positioning or sizes.

Inputs

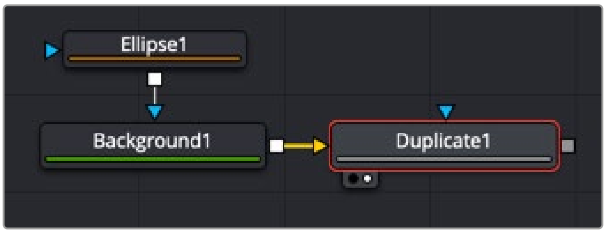
The two inputs on the Duplicate node are used to connect a 2D image and an effect mask, which can be used to limit the area where duplicated objects appear.

Input: The orange input is used for the primary 2D image that is duplicated.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the duplicated objects to appear only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The Duplicate node can be used in a variety of different ways and with a variety of different inputs. Below, to create motion graphics, a masked Background node creates a circular shape that is duplicated in the Duplicate node.



A Duplicate node used to create a repeating circular object

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Inspector

Duplicate controls

Controls Tab

The Controls tab includes all the parameters you can use to create, offset, and scale copies of the object connected to the input on the node.

Copies

Use this slider to set the number of copies made. Each copy is a copy of the last copy. So, when set to 5, the parent is copied, then the copy is copied, then the copy of the copy is copied, and so on. This allows for some interesting effects when transformations are applied to each copy using the following controls.

Time Offset

Use the Time Offset slider to offset any animations that are applied to the original image by a set amount per copy. For example, set the value to -1.0 and use a square set to rotate on the Y-axis as the source. The first copy shows the animation from a frame earlier. The second copy shows animation from a frame before that, and so forth. This can be used with great effect on textured planes, for example, where successive frames of a clip can be shown.

Center

The X and Y Center controls set the offset position applied to each copy. An X offset of 1 would offset each copy 1 unit along the X-axis from the last copy.

Pivot

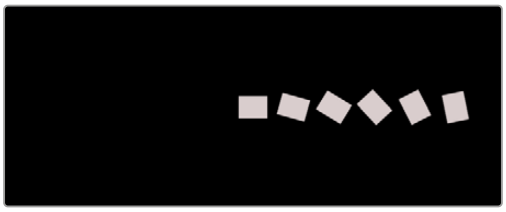
The Pivot controls determine the position of the pivot point used when changing the size, position, or angle of each copy. The pivot does not move with the original object or the duplicated array. To have the pivot follow the army, you must modify the pivot controls.

Size

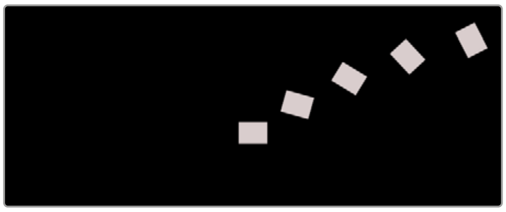
The Size control determines how much scaling to apply to each copy.

Angle

The Angle control sets the amount of Z rotation applied to each copy. The angle adjustment is linear based on the location of the pivot point.



Angle adjustment with centered pivot



Angle adjustment with offset pivot

Apply Mode

The Apply Mode setting determines the math used when blending or combining duplicated objects that overlap.

- **Normal:** The default mode uses the foreground object's Alpha channel as a mask to determine which pixels are transparent and which are not. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.
- **Screen:** Screen blends the objects based on a multiplication of their color values. The Alpha channel is ignored, and layer order becomes irrelevant. The resulting color is always lighter. Screening with black leaves the color unchanged, whereas screening with white always produces white. This effect creates a similar look to projecting several film frames onto the same surface. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.
- **Dissolve:** Dissolve mixes overlapping objects. It uses a calculated average of the objects to perform the mixture.
- **Multiply:** Multiplies the values of a color channel. This gives the appearance of darkening the object as the values are scaled from 0 to 1. White has a value of 1, so the result would be the same. Gray has a value of 0.5, so the result would be a darker object or, in other words, an object half as bright.
- **Overlay:** Overlay multiplies or screens the color values of the foreground object, depending on the color values of the object behind. Patterns or colors overlay the existing pixels while preserving the highlights and shadows of the color values of the objects behind the foreground objects. Objects behind other objects are not replaced but mixed with the front objects to reflect the original lightness or darkness of the objects behind.

- **Soft Light:** Soft Light darkens or lightens the foreground object, depending on the color values of the objects behind them. The effect is similar to shining a diffused spotlight on the image.
- **Hard Light:** Hard Light multiplies or screens the color values of the foreground object, depending on the color values of the objects behind them. The effect is similar to shining a harsh spotlight on the image.
- **Color Dodge:** Color Dodge uses the foreground object's color values to brighten the objects behind them. This is similar to the photographic practice of dodging by reducing the exposure of an area of a print.
- **Color Burn:** Color Burn uses the foreground object's color values to darken the objects behind them. This is similar to the photographic practice of burning by increasing the exposure of an area of a print.
- **Darken:** Darken looks at the color information in each channel and selects the object's foreground or background's color value, whichever is darker, as the result color. Pixels lighter than the blended colors are replaced, and pixels darker than the blended color do not change.
- **Lighten:** Lighten looks at the color information in each channel and selects the object's foreground or background's color values, whichever is lighter, as the result color value. Pixels darker than the blended color are replaced, and pixels lighter than the blended color do not change.
- **Difference:** Difference looks at the color information in each channel and subtracts the foreground object's color values from the background object's color values or the behind object's values from the foreground object's values, depending on which has the higher brightness value. Blending with white inverts the color. Blending with black produces no change.
- **Exclusion:** Exclusion creates an effect similar to but lower in contrast than the Difference mode. Blending with white inverts the base color values. Blending with black produces no change.
- **Hue:** Hue creates a result color with the luminance and saturation of the background objects color values and the hue of the foreground object's color values.
- **Saturation:** Saturation creates a result color with the luminance and hue of the base color and the saturation of the blend color.
- **Color:** Color creates a result color with the luminance of the background object's color value and the hue and saturation of the objects in the foreground. This preserves the gray levels in the image and is useful for colorizing monochrome objects.
- **Luminosity:** Luminosity creates a color using the hue and saturation of the background object and the luminance of the foreground object. This mode creates an inverse effect from that of the Color mode.

Operator

This menu is used to select the Operation mode used when the duplicate objects overlap. Changing the Operation mode changes how the overlapping objects are combined. This drop-down menu is visible only when the Apply mode is set to Normal.

The formula used to combine pixels in the Duplicate node is always (fg object * x) + (bg object * y). The different operations determine what x and y are, as shown in the description for each mode.

The Operator Modes are as follows:

- **Over:** The Over mode adds the foreground object to the background object by replacing the pixels in the background with the pixels from the Z wherever the foreground object's Alpha channel is greater than 1.

  ```
  x = 1, y = 1 – [foreground object Alpha]
  ```

- **In:** The In mode multiplies the Alpha channel of the background object against the pixels in the foreground object. The color channels of the foreground object are ignored. Only pixels from the

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foreground object are seen in the final output. This essentially clips the foreground object using the mask from the background object.

```
x = [background Alpha], y = 0
```

- **Held Out:** Held Out is essentially the opposite of the In operation. The pixels in the foreground object are multiplied against the inverted Alpha channel of the background object.

```
x = 1 – [background Alpha], y = 0
```

- **Atop:** Atop places the foreground object over the background object only where the background object has a matte.

```
x = [background Alpha], y = 1 – [foreground Alpha]
```

- **XOr:** XOr combines the foreground object with the background object wherever either the foreground or the background have a matte, but never where both have a matte.

```
x = 1 – [background Alpha], y = 1–[foreground Alpha]
```

Subtractive/Additive

This slider controls whether Fusion performs an Additive composite, a Subtractive composite, or a blend of both when the duplicate objects overlap. This slider defaults to Additive assuming the input image's Alpha channel is premultiplied (which is usually the case). If you don't understand the difference between Additive and Subtractive compositing, here's a quick explanation.

An Additive blend operation is necessary when the foreground image is premultiplied, meaning that the pixels in the color channels have been multiplied by the pixels in the Alpha channel. The result is that transparent pixels are always black since any number multiplied by 0 always equals 0. This obscures the background (by multiplying with the inverse of the foreground Alpha), and then adds the pixels from the foreground.

A Subtractive blend operation is necessary if the foreground image is not premultiplied. The compositing method is similar to an additive composite, but the foreground image is first multiplied by its Alpha, to eliminate any background pixels outside the Alpha area.

While the Additive/Subtractive option is often an either/or mode in most other applications, the Duplicate node lets you blend between the Additive and Subtractive versions of the compositing operation. This can be useful for dealing with problem composites with bright or dark edges.

For example, using Subtractive merging on a premultiplied image may result in darker edges, whereas using Additive merging with a non-premultiplied image causes any non-black area outside the foreground's Alpha to be added to the result, thereby lightening the edges. By blending between Additive and Subtractive, you can tweak the edge brightness to be just right for your situation.

Gain

The Gain RGB controls multiply the values of the image channel linearly. All pixels are multiplied by the same factor, but the effect is larger on bright pixels and smaller on dark pixels. Black pixels are not changed since multiplying any number times 0 always equals 0.

Alpha Gain linearly scales the Alpha channel values of objects in front. This effectively reduces the amount that the objects in the background are obscured, thus brightening the overall result. When the Subtractive/Additive slider is set to Additive with Alpha Gain set to 0.0, the foreground pixels are simply added to the background.

When Subtractive/Additive slider is set to Subtractive, this controls the density of the composite, similarly to Blend.

All Gain values will compound based on the number of duplications.

Blur

Adds a blurring effect to the duplicated layers.

- **Lock Blur:** Locks the X and Y Blur sliders together for symmetrical blurring. This is enabled by default. When the Lock Blur control is deselected, independent control over each axis is provided.
- **Blur:** Sets the amount of blur applied to the duplicated layers in the tool. The Blur amount will not compound based on the number of duplications.
- **Glow:** Adds a glow effect to the blur of the duplicated layers.
- **Blend:** The Blend slider determines the percentage of the affected image that is mixed with original image. It blends in more of the original image as the value gets closer to 0.
- **RGBA Scale:** Allows adjusting the strength of the individual Red, Green, Blue, and Alpha channels to the blur of the duplicated layers.

Burn In

The Burn In control adjusts the amount of Alpha used to darken the objects that fall behind other objects, without affecting the amount of foreground objects added. At 0.0, the blending behaves like a straight Alpha blend, in contrast to a setting of 1.0 where the objects in the front are effectively added on to the objects in the back (after Alpha multiplication if in Subtractive mode). This gives the effect of the foreground objects brightening the objects in the back, as with Alpha Gain. In fact, for Additive blends, increasing the Burn In gives an identical result to decreasing Alpha Gain.

Blend

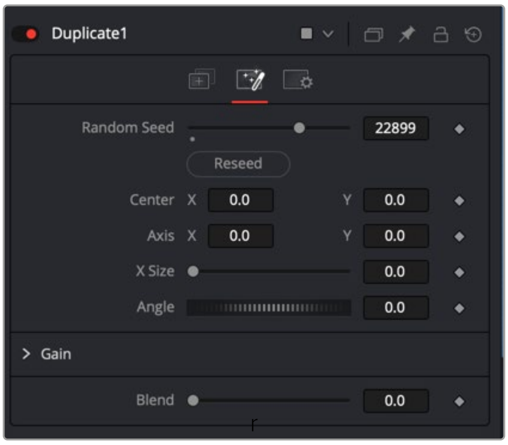
This blend control is different from the Blend slider in the Common Settings tab. Changes made to this control apply the blend between objects. The Blend slider fades the results of the last object first, the penultimate after that, and so on. The blending is divided between 0 and 1, with 1 being all objects are fully opaque and 0 being only the original object showing.

Merge Under

This checkbox reverses the layer order of the duplicated elements, making the last copy the bottommost layer and the first copy the topmost layer.

Jitter Tab

The options in the Jitter tab allow you to randomize the position, rotation, size, and color of all the copies created in the Controls tab.



Duplicate Jitter tab

Random Seed

The Random Seed slider and Reseed button are used to generate a random starting point for the amount of jitter applied to the duplicated objects. Two Duplicate nodes with identical settings but different random seeds produce two completely different results.

Center X and Y

Use these two controls to adjust the amount of variation in the X and Y position of the duplicated objects.

Axis X and Y

Use these two controls to adjust the amount of variation in the rotational pivot center of the duplicated objects. This affects only the additional jitter rotation, not the rotation produced by the Rotation settings in the Controls tab.

X Size

Use this control to adjust the amount of variation in the Scale of the duplicated objects.

Angle

Use this dial to adjust the amount of variation in the Z rotation of the duplicated objects.

Gain

The Gain RGBA controls randomly multiply the values of the image channel linearly.

Blend

Changes made to this control randomize the blend between objects.

Common Controls

Settings Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

Highlight [HIL]

The Highlight node

Highlight Node Introduction

The Highlight filter creates star-shaped highlights or glints in bright regions of the image, similar to a lens star filter effect.

Inputs

There are three Inputs on the Highlight node: one for the image, one for the effects mask, and another for a highlight mask.

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Input: The orange input is used for the primary 2D image that gets the highlight applied.

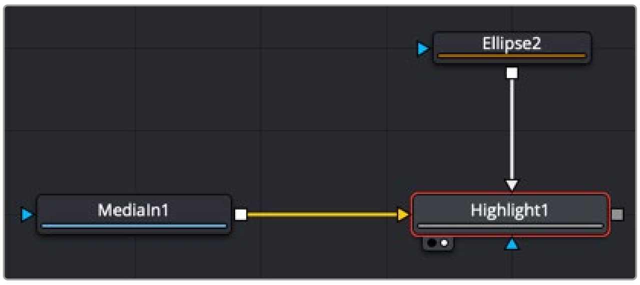
Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input restricts the highlight to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

Highlight Mask: The Highlight node supports pre-masking using the white highlight mask input. The image is filtered before the highlight is applied. The highlight is then merged back over the original image. Unlike regular effect masks, it does not crop off highlights from source pixels when the highlight extends past the edges of the mask.

Highlight masks are identical to effects masks in every other respect.

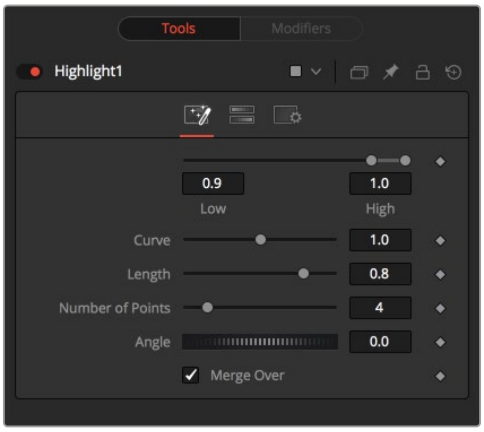
Basic Node Setup

The Highlight node below is used to create glint-type highlights on an incoming image. The highlight mask is used to limit the area where the effect is applied.



A Highlight node applied to an image, with a highlight mask limiting the area of the effect

Inspector



Highlight controls

Controls Tab

The Controls tab includes parameters for the highlight style except for color, which is handled in the Color Scale tab.

Low and High

This range control designates the range of Luminance values in the image that generates highlights. Values less than the Low value do not receive highlights. Values above the High value receive the full highlight effect.

Curve

The Curve value changes the drop-off over the length of the highlight. Higher values cause the brightness of the flares to drop off closer to the center of the highlight, whereas lower values drop off farther from the center.

Length

This designates the length of the flares from the highlight.

Number of Points

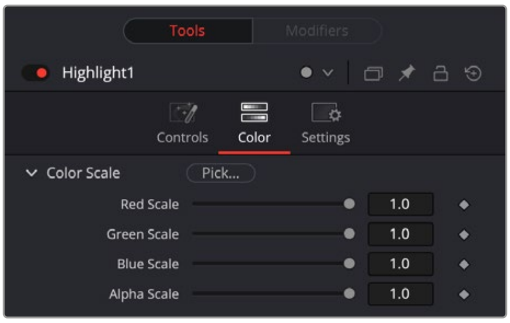
This determines the number of flares emanating from the highlight.

Angle

Use this control to rotate the highlights.

Merge Over

When enabled, the effect is overlaid on the original image. When disabled, the output is the highlights only. This is useful for downstream color correction of the highlights.



Highlight Color Scale controls

Color Scale Tab

The Color Scale tab controls the color of the highlight.

By click and holding on the Pick button, then dragging the pointer over the viewer, you can select a specific color from the image.

Red, Green, and Blue Scale

Moving the sliders of one or all of these channels down changes the falloff color of the highlight.

Alpha Scale

Moving the Alpha slider down makes highlight falloff more transparent.

Common Controls

Setting Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

Hot Spot [Hot]

The Hot Spot node

Hot Spot Node Introduction

The Hot Spot node is used to create lens flare, spotlight, and burn/dodge effects of various types.

In the real world, lens flares occur when extremely bright light sources in the scene by the reflections are reflected off elements inside the lens of the camera. One might see lens flares in a shot when viewing a strong light source through a camera lens, like the sun or another bright star.

Inputs

There are three inputs on the Hot Spot node: one for the image, one for the effects mask, and another for an Occlusion image.

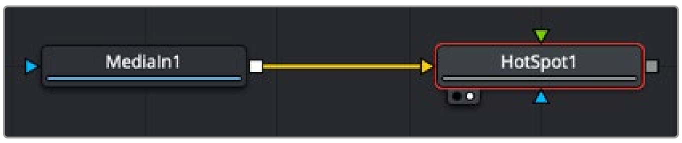
Input: The required orange input is used for the primary 2D image that gets the hot spot applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input restricts the hot spot to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

Occlusion: The green Occlusion input accepts an image to provide the occlusion matte. The matte is used to block the hot spot, causing it to “wink.” The white pixels in the image occlude the hot spot. Gray pixels partially suppress the hot spot.

Basic Node Setup

The Hot Spot node is not a stand-alone generator, so it must have an image input that gets the hot spot applied.



Hot Spot node applied to an image

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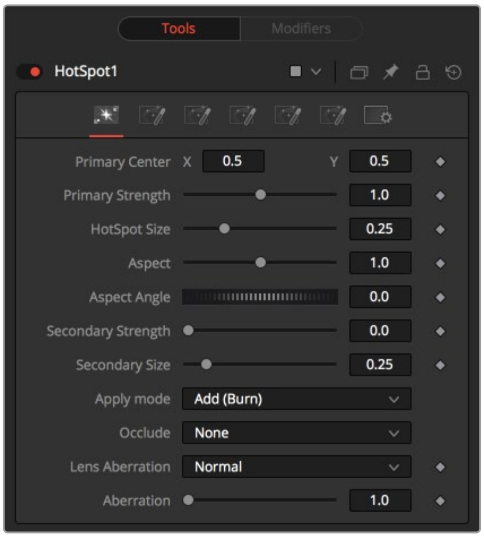
Inspector



Hot Spot controls

Hot Spot Tab

The Hotspot tab is used to control the primary and secondary hot spots. You can adjust their position, size, strength, angle, and apply mode.

Primary Center X and Y

This is the position of the primary hot spot within the scene. Secondary lens elements and reflections are positioned relative to the position of the primary hot spot.

Primary Strength

This control determines the brightness of the primary hot spot.

Hot Spot Size

This control determines the diameter of the primary hot spot. A value of 1.0 represents a circle the full width of the image.

Aspect

This controls the aspect of the spot. A value of 1.0 produces a perfectly circular hot spot. Values above 1.0 elongate the circle horizontally, and values below 1.0 elongate the circle vertically.

Aspect Angle

This control can be used to rotate the primary hot spot.

Secondary Strength

This control determines the strength, which is to say the brightness, of the secondary hot spot. The secondary hot spot is a reflection of the primary hot spot. It is always positioned on the opposite side of the image from the primary hot spot.

Secondary Size
This determines the size of the secondary hot spot.

Apply Mode
This control determines how the hot spot affects the underlying image.

- **Add (Burn):** This causes the spots created to brighten the image.
- **Subtract (Dodge):** This causes the spots created to dim the image.
- **Multiply (Spotlight):** This causes the spots created to isolate a portion of the image with light and to darken the remainder of the image.

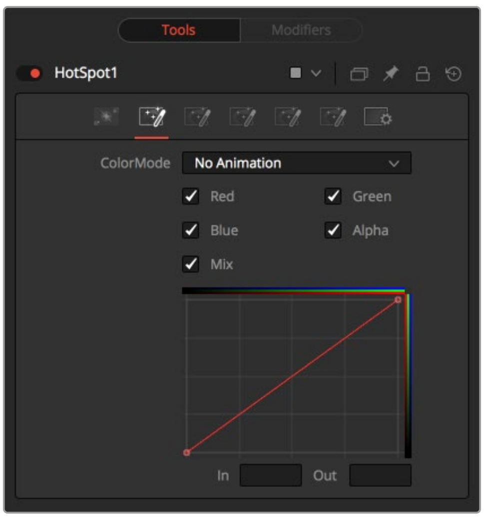
Occlude
This menu is used to select which channel of the image connected to the Hot Spot node's Occlusion input is used to provide the occlusion matte. Occlusion can be controlled from Alpha or R, G, or B channels of any image connected to the Occlusion input on the node's tile.

Lens Aberration
Aberration changes the shape and behavior of the primary and secondary hot spots.

- **In and Out Modes:** Elongates the shape of the hot spot into a flare. The hot spot stretches toward the center when set to In mode and stretches toward the corners when set to Out mode.
- **Flare In and Flare Out Modes:** This option is a lens distortion effect that is controlled by the movement of the lens effect. Flare In causes the effect to become more severe, the closer the hot spot gets to the center. Flare Out causes the effect to increase as the hot spot gets closer to the edges of the image.
- **Lens:** This mode emulates a round, ringed lens effect.

Aberration
The Aberration slider controls the overall strength of the lens aberration effect.

Color Tab
The Color tab is used to modify the color of the primary and secondary hot spots.



Hot Spot color controls

Color Mode

This menu allows you to choose between animated or static color modifications using the small curves editor in the Inspector.

- **None:** The default None setting retains a static curve adjustment for the entire range.
- **Animated Points:** This setting allows the color curves in the spline area to be animated over time. Once this option is selected, moving to the desired frame and making a change in the Spline Editor sets a keyframe.
- **Dissolve mode:** Dissolve mode is mostly obsolete and is included for compatibility reasons only.

Color Channel and Mix

When selected, these checkboxes enable the editing of the chosen splines in the small Inspector Spline Editor. The Mix checkbox enables the Mix Spline, which is used to determine the influence of the controls that the Radial tab has along the radius of the hot spot.

Red, Green, Blue, and Alpha Splines

The Spline Window shows the curves for the individual channels. It is a miniature Spline Editor. The Red, Green, Blue, and Alpha splines are used to adjust the color of the spotlight along the radius of the hot spot.

The vertical axis represents the intensity or strength of the color channel. The horizontal axis represents the hot spot position along the radius, from the left outside edge to the inside right edge.

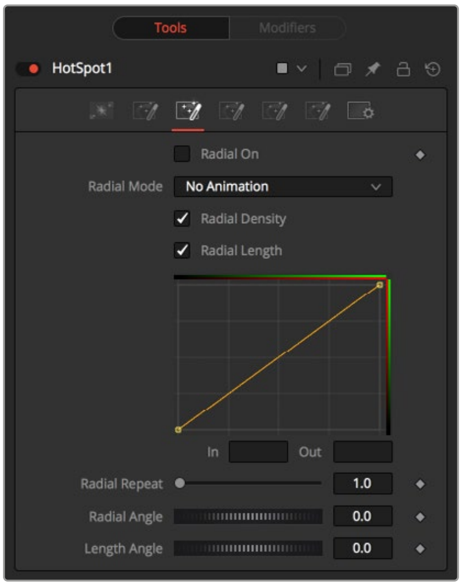
The default curve indicates that the red, green, blue, and Alpha channels all have a linear falloff.

Mix Spline

The Mix spline is used to determine the influence that the Radial controls have along the radius of the hot spot. The horizontal axis represents the position along the circle's circumference, with 0 being 0 degrees and 1.0 being 360 degrees. The vertical axis represents the amount of the radial hot spot to blend with the color hot spot. A value of 0 is all radial hot spot, while a value of 1.0 is all color hot spot.

NOTE: Right-clicking in the LUT displays a contextual menu with options related to modifying spline curves.

For more information on the LUT Editor, *see Chapter 69, "Using Viewers."* in the DaVinci Resolve Reference Manual, or Chapter 7 in the Fusion Reference Manual.



Hot Spot Radial tab

Radial Tab

Radial On

This control enables the Radial splines. Otherwise, the radial matte created by the splines is not applied to the hot spot, and the Mix spline in the color controls does not affect the hot spot.

Radial Mode

Similar to the Color mode menu, this menu allows you to choose between animated or static radial hot spot modifications using the small curves editor in the Inspector.

- **No Animation:** The default setting retains a static curve adjustment for the entire range.
- **Animated Points:** This setting allows the radial curves in the spline area to be animated over time. Once this option is selected, moving to the desired frame and making a change in the Spline Editor sets a keyframe.

The Interpolated Values option is mostly obsolete and is included for compatibility reasons only.

Radial Length and Radial Density Splines

The Spline window shows curves for the Length and Density of the hot spot. It is a miniature Spline Editor. The key to these splines is realizing that the horizontal axis in Inspector's Spline Editor represents a position around the circumference of the hot spot. A value of 0.0 is 0 degrees, and 1.0 is 360 degrees. With that in mind, the length determines the radius of light making up the hot spot along the circumference. The density represents how bright the light is along the circumference.

Radial Repeat

This control repeats the effect of the radial splines by x number of times. For example, a repeat of 2.0 causes the spline to take effect between 0 and 180 degrees instead of 0 and 360, repeating the spline between 180 and 360.

Length Angle

This control rotates the effect of the Radial Length spline around the circumference of the hot spot.

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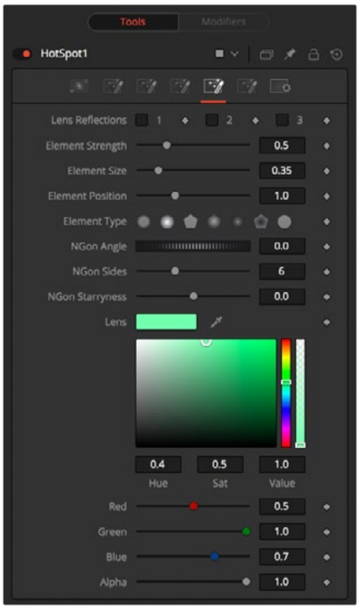
Density Angle

This control rotates the effect of the Radial Density spline around the circumference of the hot spot.

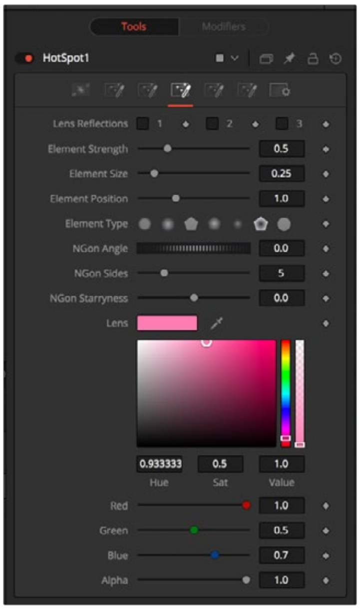
NOTE: Right-clicking in the spline area displays a contextual menu containing options related to modifying spline curves.

A complete description of LUT Editor controls and options, *see Chapter 107, "LUT Nodes,"* in DaVinci Resolve Reference Manual and Chapter 47 in the Fusion Reference Manual.

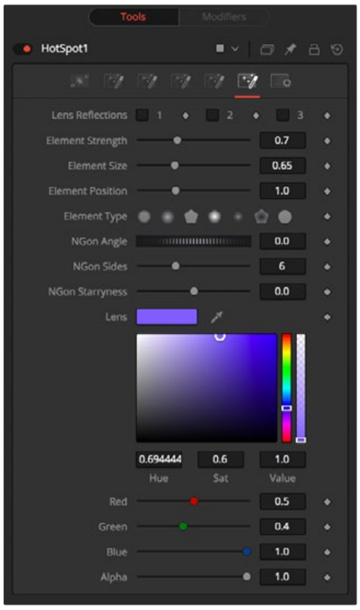
L1, L2, and L3 Tab



L1 tab



L2 tab



L3 tab

Lens Reflect Tabs

The three Lens Reflect tabs are used to enable and design additional lens flare elements beyond the primary and secondary hot spots.

Lens Reflect 1-3

Each of these three checkboxes enables a pair of lens reflection elements that you can modify using the controls in this tab. The parameters affect all the enabled Lens reflection elements in this tab.

Element Strength

This determines the brightness of element reflections.

Element Size

This determines the size of element reflections.

Element Position

This determines the distance of element reflections from the axis. The axis is calculated as a line between the hot spot position and the center of the image.

Element Type

Use this group of buttons to choose the shape and density of the element reflections. The presets available are described below.

- **Circular:** This creates slightly soft-edged circular shaped reflections.
- **Soft Circular:** This creates very soft-edged circular shaped reflections.
- **Circle:** This creates a hard-edged circle shape.
- **NGon Solid:** This creates a filled polygon with a variable number of sides.
- **NGon Star:** This creates a very soft-edged star shape with a variable number of sides.
- **NGon Shaded Out:** This creates soft-edged circular shapes.
- **NGon Shaded In:** This creates a polygon with a variable number of sides, which has a very soft reversed (dark center, bright radius) circle.
- **NGon Angle:** This control is used to determine the angle of the NGon shapes.
- **NGon Sides:** This control is used to determine the number of sides used when the Element Type is set to Ngon Star, Ngon Shaded Out, and Ngon Shaded In.
- **NGon Starriness:** This control is used to bend polygons into star shapes. The higher the value, the more star-like the shape.

Lens Color Controls

These controls determine the color of the lens that affects the colors of the reflections. To choose a lens color, pick one from a displayed image or enter RGBA values using the sliders or input boxes.

Common Controls

Settings Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

Object Removal [ORm]

The Object Removal node

Object Removal Node Introduction

Object Removal uses the DaVinci Neural Engine to attempt to remove an object in the frame as automatically as possible. This node works best when removing a moving object that passes over a temporally stable background, or dirt on the lens of a shot where the camera is in motion. Smaller objects get better results than larger objects, but your results really depend on the footage.

Inputs

You can add masks and clean plates to the Object Removal node.

Input: The orange input is used for the primary footage you wish to remove an object from.

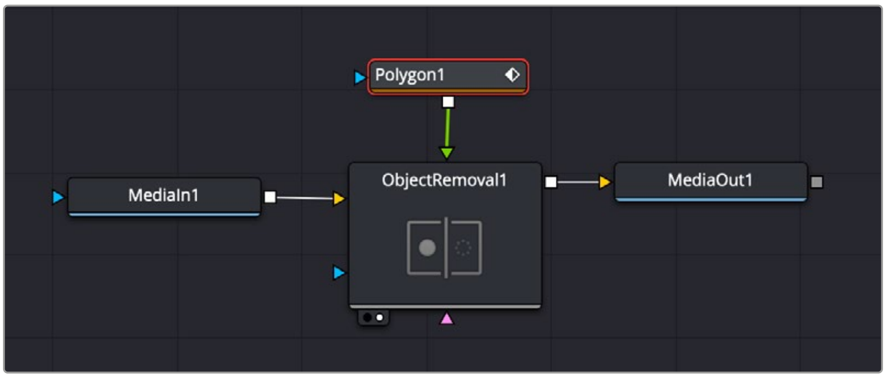
Clean Plate: The magenta input accepts a clean plate image source, if selected in the controls.

Mask: The green input accepts a mask shape (i.e., polygon, ellipse, rectangle, etc.) that is drawn around the object you want to remove from the scene.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the filter to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

Connect the media you wish to remove the object from to the yellow Input, and then either a polygon shape to the green Mask input, or a frame without the object to the clean plate input.



The Object Removal tool node structure. You will need either a polygon or another shape attached to the green Mask input, or a clean plate attached to the pink clean plate input.

To setup the Object Removal:

1. Draw a polygon around the object that you want to remove from the scene in the Polygon node.
2. Track the polygon either manually, or using a tracker, to make sure that the object you want to remove is always inside its boundaries through the duration of the clip.
3. Press the Scene Analysis button in the Object Removal tool controls.

These steps should accommodate most simple object removal tasks, but if there are issues, you can adjust the parameters of the tool in the Inspector. The Object Removal tool is not appropriate for all scenes and works best with small objects that move through the frame, not in front of or behind other objects.

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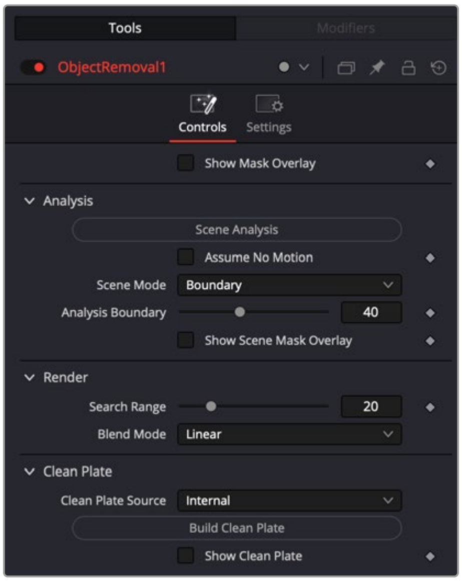
Drawing and tracking a polygon around the cyclist, then pressing the Scene Analysis button in the Object Removal controls, removes the cyclist from the background.

Inspector



The Object Removal controls help you remove unwanted objects from an image.

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Controls

The Object Removal controls let you adjust the parameters of the object analysis.

- **Show Mask Overlay:** Lets you see the actual mask used, highlighted in red, for the Object Removal.

Analysis

- **Scene Analysis:** Press this button to analyze the clip using the selected parameters and replace the object with the appropriate background. You will need to press this button again if you change any of the parameters in this section.
- **Assume No Motion:** If the camera has been locked off for the shot while the object moves through it, check this box to greatly simplify the background analysis.
- **Scene Mode:** Provides different methods of analyzing the scene, improving the analysis of how the area that needs to be replaced moves to best determine how to fill the hole left by the object being removed.

  Background: Analyzes the entire image except for the object region.

  Boundary: Analyzes the boundary area surrounding the object region.

  Object: For analyzing an object that moves with the background, like a sticker that's on a window while the camera moves.

- **Analysis Boundary:** Sets the object's boundary size for analysis.
- **Show Scene Mask Overlay:** Toggles the scene mask overlay on or off, letting you see what's behind the mask.

Render

- **Search Range:** If you notice, while playing through the analyzed result, that the object removal mask has gray fringing on some frames, you can try adjusting the "Search Range" slider, which is the distance, in frames, from the current frame that the Object Removal plugin is searching for replacement image detail. For example, if the Search Range is 20, it searches +/-20 frames from the current location, or 40 frames total. The allowance of 10 frames means we look at every fourth frame. You will generally get the best results for the smallest range that gives an acceptable result.
- **Blend Mode:** If the patch is successfully filled, but the result isn't blending well with the background, you can try changing the Blend mode. The default is Linear, which is a simple cloning operation, but you can also choose Adaptive Blend, which can provide better results except in certain situations where the edges of the replacement patch have a different color or brightness than the background.

Clean Plate

- **Clean Plate Source:** Lets you set the source of the clean plate. Options are Gray Image, which is effectively no source; Internal, which takes a "best guess" approach to generating a background with which to fill the frame and integrates this with frames that could be successfully filled in; and External, which lets you connect your own plate to the pink input from another node.
- **Build Clean Plate:** Executes the building of a new clean plate background based on the Clean Plate Source parameter set.
- **Show Clean Plate:** Checking this box lets you see behind the mask to the clean plate behind it.

Pseudo Color [PsCl]

The Pseudo Color node

Pseudo Color Node Introduction

The Pseudo Color node provides the ability to produce variations of an image's color based on waveforms generated by the node's controls. Static or animated variances of the original image can be produced.

Inputs

There are two Inputs on the Pseudo Color node: one for an image and one for an effects mask.

Input: The orange input is used for the primary 2D image that gets its color modified.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input restricts the pseudo color to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The Pseudo Color node is not a stand-alone generator, so it must have an image input that it uses to generate variations in colors.

Pseudo Color node applied to an image

Inspector

Pseudo Color RGBA controls

Red/Green/Blue/Alpha Tabs

The node's controls are separated into four identical tabs, one for each of the RGBA color channels.

Color Checkbox

When enabled, the Pseudo Color node affects this color channel.

Wrap

When enabled, waveform values that exceed allowable parameter values are wrapped to the opposite extreme.

High and Low

High and Low determine the range to be affected by the node in a specific color channel.

Soft Edge

This slider determines the soft edge of color transition.

Waveform

This selects the type of waveform to be created by the generator. Four waveforms are available: Sine, Triangle, Sawtooth, and Square.

Frequency

This controls the frequency of the waveform selected. Higher values increase the number of occurrences of the variances.

Phase

This modifies the Phase of the waveform. Animating this control produces color cycling effects.

Mean

This determines the level of the waveform selected. Higher values increase the overall brightness of the channel until the allowed maximum is reached.

Amplitude

Amplitude increases or decreases the overall power of the waveform.

Common Controls

Settings Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

Rays [CIR]

The Rays node

Rays Node Introduction

Rays is a modified zoom blur effect that radiates through an object from a specified point.

Inputs

There are two inputs on the Rays node: one for the image and one for the effects mask.

Input: The orange input is used for the primary 2D image that gets the rays applied to it.

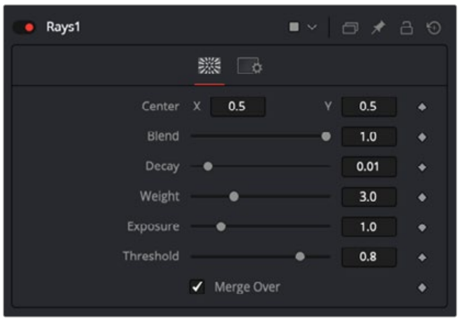
Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input restricts the rays to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

Basic Node Setup

The Rays node works best when the image or graphic connected to the orange input includes an Alpha channel from which the rays emit.

Rays node set up to emit from a line of text

Inspector



Rays node controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the rays.

Center X and Y

This coordinate control and related viewer crosshair set the center point for the light source.

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Blend

Sets the percentage of the original image that's blended with the light rays.

Decay

Sets the length of the light rays.

Weight

Sets the falloff of the light rays.

Exposure

Sets the intensity level of the light rays.

Threshold

Sets the luminance limit at which the light rays are produced.

Common Controls

Settings Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in the "The Common Controls" section at the end of this chapter.

Shadow [SH]

The Shadow node

Shadow Node Introduction

Shadow is a versatile node used in the creation of a drop shadow, based on the Alpha channel in an image. Optionally, a second image can be used as a depth matte to distort the shadow based on the varying depth in a background image.

Input

The three inputs on the Shadow node are used to connect a 2D image that causes the shadow. A depth map input and an effect mask can be used to limit the area where trails appear. Typically, the output of the shadow is then merged over the actual background in the composite.

Input: The orange input is used for the primary 2D image with Alpha channel that is the source of the shadow.

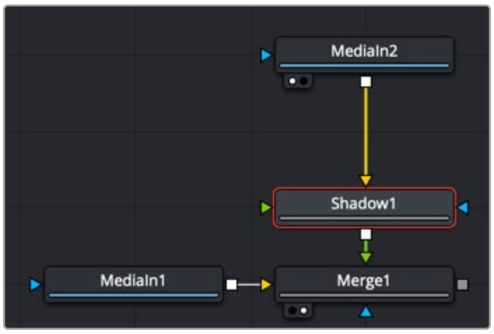
Depth: The green Depth map input takes a 2D image as its input and extracts a depth matte from a selected channel. The light Position and Distance controls can then be used to modify the appearance of the shadow based on depth.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the area where the shadow appears. An effects mask is applied to the tool after the tool is processed.

> **NOTE:** The Shadow node is designed to create simple 2D drop shadows. Use a Spot Light node and an Image Plane 3D node for full 3D shadow casting.

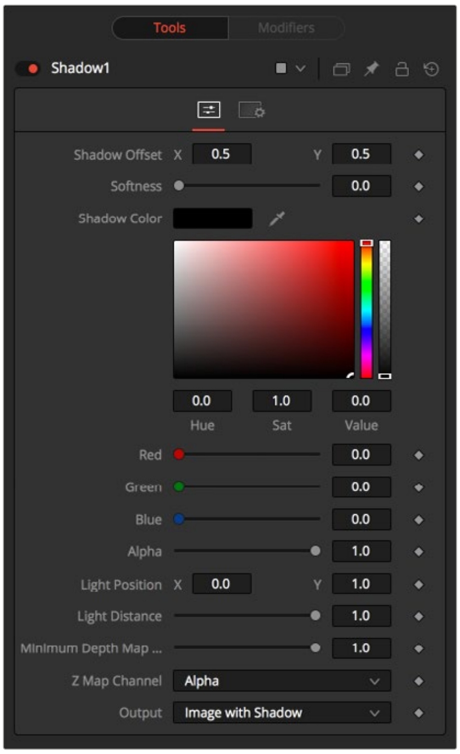
Basic Node Setup

Below, the Shadow node uses the output of an image with Alpha and connects to the foreground of a Merge. The shadow is shown over the background input to the Merge.



A Shadow node is generated from the MediaIn2 and shown over MediaIn1

Inspector



Shadow node controls

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Controls Tab

The Controls tab contains all the primary controls necessary for customizing the shadow appearance.

Shadow Offset

This control sets the X and Y position of the shadow. When the Shadow node is selected, you can also adjust the position of the Shadow Offset using the crosshair in the viewer.

Softness

Softness controls how blurry the shadow's edges appear.

Shadow Color

Use this control to select the color of the shadow. The most realistic shadows are usually not totally black and razor sharp.

Light Position

This control sets the position of the light relative to the shadow-casting object. The Light Position is only taken into consideration when the Light Distance slider is not set to infinity (1.0).

Light Distance

This slider varies the apparent distance of the light between infinity (1.0) and zero distance from the shadow-casting object. The advantage of setting the Light Distance is that the resulting shadow is more realistic-looking, with the further parts of the shadow being longer than those that are closer.

Minimum Depth Map Light Distance

This control is active when an image is connected to the shadow's Depth Map input. The slider is used to control the amount that the depth map contributes to the Light Distance. Dark areas of a depth map make the shadow deeper. White areas bring it closer to the camera.

Z Map Channel

This menu is used to select which color channel of the image connected to the node's Depth Map input is used to create the shadow's depth map. Selections exist for the RGB and A, Luminance, and Z-buffer channels.

Output

This menu determines if the output image contains the image with shadow applied or the shadow only.

The shadow only method is useful when color correction, perspective, or other effects need to be applied to the resulting shadow before it is merged back with the object.

Common Controls

Settings Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

Trails [TRLS]

The Trails node

Trails Node Introduction

The Trails node is used to create a ghost-like after-trail of the image. This creates an interesting effect when applied to moving images with an Alpha channel. Unlike a directional blur, only the preceding motion of an image is displayed as part of the effect. Since the trail effect is based on an image buffer, it requires you to play or activate the pre-roll for some number of frames before you see the effect.

Input

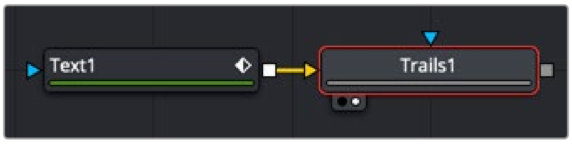
The two inputs on the Trails node are used to connect a 2D image and an effect mask that can be used to limit the area where trails appear.

Input: The orange input is used for the primary 2D image that receives the trails applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the area where the trails effect appears. An effects mask is applied to the tool after the tool is processed.

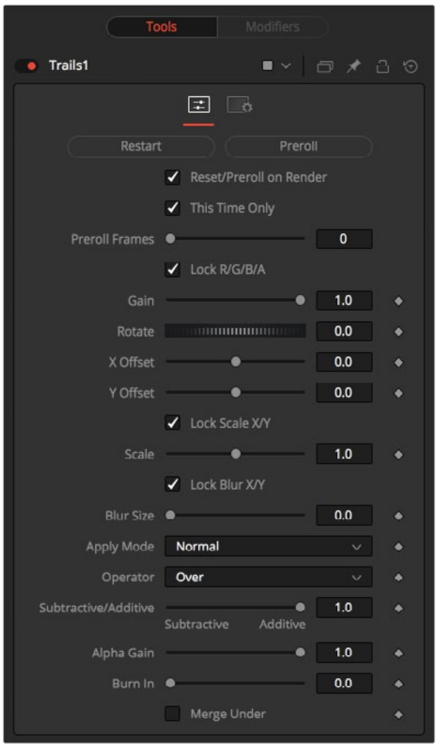
Basic Node Setup

The output of an animated Text node is connected to the input of the Trails node. Trails are generated based on the motion of the text. The Reset button must be pressed in the Inspector between each preview, or the trails will accumulate.



A Trails node generates trails for the animation in the Text node

Inspector



Trails node controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the trails.

Restart

This control clears the image buffer and displays a clean frame, without any of the ghosting effects.

Preroll

This makes the Trails node pre-render the effect by the number of frames on the slider.

Reset/Preroll on Render

When this checkbox is enabled, the Trails node resets itself when a preview or final render is initiated. It pre-rolls the designated number of frames.

This Time Only

Selecting this checkbox makes the pre-roll use this current frame only and not the previous frames.

Preroll Frames

This determines the number of frames to pre-roll.

Lock RGBA

When selected, this checkbox allows the Gain of the color channels to be controlled independently. This allows for tinting of the Trails effect.

Gain

The Gain control affects the overall intensity and brightness of the image in the buffer. Lower values in this parameter create a much shorter, fainter trail, whereas higher values create a longer, more solid trail.

Rotate

The Rotate control rotates the image in the buffer before the current frame is merged into the effect. The offset is compounded between each element of the trail. This is different than each element of the trail rotating on its pivot point. The pivot remains over the original object.

Offset X/Y

These controls offset the image in the buffer before the current frame is merged into the effect. Control is given over each axis independently. The offset is compounded between each element of the trail.

Lock Scale X/Y

When selected, this checkbox allows the X- and Y-axis scaling of the image buffer to be manipulated separately for each axis.

Scale

The Scale control resizes the image in the buffer before the current frame is merged into the effect. The size is compounded between each element of the trail.

Lock Blur X/Y

When selected, this checkbox allows the blurring of the image buffer to be controlled separately for each axis.

Blur Size

The Blur Size control applies a blur to the trails in the buffer before the current frame is merged into the effect. The blur is compounded between each element of the trail.

Apply Mode

The Apply Mode setting determines the math used when blending or combining the trailing objects that overlap.

- **Normal:** The default mode uses the foreground object's Alpha channel as a mask to determine which pixels are transparent and which are not. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.
- **Screen:** Screen blends the objects based on a multiplication of their color values. The Alpha channel is ignored, and layer order becomes irrelevant. The resulting color is always lighter. Screening with black leaves the color unchanged, whereas screening with white always produces white. This effect creates a similar look to projecting several film frames onto the same surface. When this is active, another menu shows possible operations, including Over, In, Held Out, Atop, and XOr.
- **Dissolve:** Dissolve mixes overlapping objects. It uses a calculated average of the objects to perform the mixture.
- **Multiply:** Multiplies the values of a color channel. This gives the appearance of darkening the object as the values are scaled from 0 to 1. White has a value of 1, so the result would be the same. Gray has a value of 0.5, so the result would be a darker object or, in other words, an object half as bright.
- **Overlay:** Overlay multiplies or screens the color values of the foreground object, depending on the color values of the background object. Patterns or colors overlay the existing pixels while preserving the highlights and shadows of the color values of the objects behind the foreground objects. The objects behind the foreground objects are not replaced but mixed with the foreground objects to reflect the original lightness or darkness of the background objects.
- **Soft Light:** Soft Light darkens or lightens the foreground object, depending on the color values of the objects behind them. The effect is similar to shining a diffused spotlight on the image.

- **Hard Light:** Hard Light multiplies or screens the color values of the foreground object, depending on the color values of the objects behind them. The effect is similar to shining a harsh spotlight on the image.
- **Color Dodge:** Color Dodge uses the foreground object's color values to brighten the objects behind them. This is similar to the photographic practice of dodging by reducing the exposure of an area of a print.
- **Color Burn:** Color Burn uses the foreground object's color values to darken the objects behind them. This is similar to the photographic practice of burning by increasing the exposure of an area of a print.
- **Darken:** Darken looks at the color information in each channel and selects the color value from the object in front or behind, whichever is darker. Pixels lighter than the blended colors are replaced, and pixels darker than the blended color do not change.
- **Lighten:** Lighten looks at the color information in each channel and selects the color value from the object in front or behind, whichever is lighter. Pixels darker than the blended color are replaced, and pixels lighter than the blended color do not change.
- **Difference:** Difference looks at the color information in each channel and subtracts the foreground object's color values from the background object's color values or vice versa, depending on which has the higher brightness value. Blending with white inverts the color. Blending with black produces no change.
- **Exclusion:** Exclusion creates an effect similar to but lower in contrast than the Difference mode. Blending with white inverts the base color values. Blending with black produces no change.
- **Hue:** Hue creates color with the luminance and saturation of the background object's color and the hue of the foreground object's color.
- **Saturation:** Saturation creates color with the luminance and hue of the base color and the saturation of the blend color.
- **Color:** Color creates color with the luminance of the background object's color and the hue and saturation of the object in front. This preserves the gray levels in the image and is useful for colorizing monochrome objects.
- **Luminosity:** Luminosity creates color with the hue and saturation of the background object's color and the luminance of the foreground object's color. This mode creates an inverse effect from that of the Color mode.

Operator

This menu is used to select the Operation mode used when the trailing objects overlap. Changing the Operation mode changes how the overlapping objects are combined to produce a result. This drop-down menu is visible only when the Apply mode is set to Normal.

The formula used to combine pixels in the trails node is always (fg object * x) + (bg object * y). The different operations determine what x and y are, as shown in the description for each mode.

The Operator Modes are as follows:

- **Over:** The Over mode adds the foreground object to the background object by replacing the pixels in the background with the pixels from the Z wherever the foreground object's Alpha channel is greater than 1.

  ```
  x = 1, y = 1 – [foreground object Alpha]
  ```

- **In:** The In mode multiplies the Alpha channel of the background object against the pixels in the foreground object. The color channels of the foreground object are ignored. Only pixels from the foreground object are seen in the final output. This essentially clips the foreground object using the mask from the background object.

  ```
  x = [background Alpha], y = 0
  ```

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- **Held Out:** Held Out is essentially the opposite of the In operation. The pixels in the foreground object are multiplied against the inverted Alpha channel of the background object.

  `x = 1 – [background Alpha], y = 0`

- **Atop:** Atop places the foreground object over the background object only where the background object has a matte.

  `x = [background Alpha], y = 1 – [foreground Alpha]`

- **XOr:** XOr combines the foreground object with the background object wherever either the foreground or the background have a matte, but never where both have a matte.

  `x = 1 – [background Alpha], y = 1 – [foreground Alpha]`

Subtractive/Additive

This slider controls whether Fusion performs an Additive composite, a Subtractive composite, or a blend of both when the trailing objects overlap. This slider defaults to Additive assuming the input image's Alpha channel is premultiplied (which is usually the case). If you don't understand the difference between Additive and Subtractive compositing, below is a quick explanation.

> **NOTE:** An Additive blend operation is necessary when the foreground image is premultiplied, meaning that the pixels in the color channels have been multiplied by the pixels in the Alpha channel. The result is that transparent pixels are always black since any number multiplied by 0 always equals 0. This obscures the background (by multiplying with the inverse of the foreground Alpha), and then adds the pixels from the foreground.
>
> A Subtractive blend operation is necessary if the foreground image is not premultiplied. The compositing method is similar to an additive composite, but the foreground image is first multiplied by its Alpha, to eliminate any background pixels outside the Alpha area.
>
> Although the Additive/Subtractive option is often an either/or checkbox in other software, the Trails node lets you blend between the Additive and Subtractive versions of the compositing operation. This can be useful when dealing with problem edges that are too bright or too dark.

For example, using Subtractive merging on a premultiplied image may result in darker edges, whereas using Additive merging with a non-premultiplied image causes any non-black area outside the foreground's Alpha to be added to the result, thereby lightening the edges. By blending between Additive and Subtractive, you can tweak the edge brightness to be just right for your situation.

Alpha Gain

Alpha Gain linearly scales the Alpha channel values of the trailing objects in front. This effectively reduces the amount that the trailing objects in the background are obscured, thus brightening the overall result. When the Subtractive/Additive slider is set to Additive with Alpha Gain set to 0.0, the foreground pixels are added to the background.

When the Subtractive/Additive slider is set to Subtractive, this controls the density of the composite, similar to Blend.

Burn In

The Burn In control adjusts the amount of Alpha used to darken the objects that trail under other objects, without affecting the amount of foreground objects added. At 0.0, the blending behaves like a straight Alpha blend. At 1.0, the objects in the front are effectively added onto the objects in the back (after Alpha multiplication if in Subtractive mode). This gives the effect of the foreground objects

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brightening the objects in the back, as with Alpha Gain. In fact, for Additive blends, increasing the Burn In gives an identical result to decreasing Alpha Gain.

Merge Under

When enabled, the current image is placed under the generated trail, rather than the usual, over top operation. The layer order of the trailing elements is also reversed, making the last trail the topmost layer.

Common Controls

Settings Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

TV [TV]

The TV node

TV Node Introduction

The TV node is a simple node designed to mimic some of the typical flaws seen in analog television broadcasts and screens. This Fusion-specific node is mostly obsolete when using DaVinci Resolve because of the more advanced Analog Damage ResolveFX.

Input

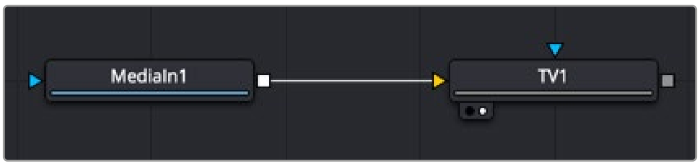
The two inputs on the TV node are used to connect a 2D image and an effect mask, which can be used to limit the area where the TV effect appears.

Input: The orange input is used for the primary 2D image that gets the TV distortion applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the area where the TV effect to appears. An effects mask is applied to the tool after the tool is processed.

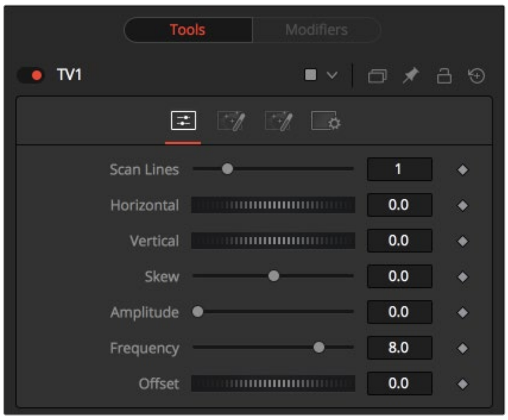
Basic Node Setup

The output of an image is connected to the input of the TV node. The style of TV interference is then customized using the controls in the Inspector.



The TV node simulates TV-style flaws in the image connected to the orange input

Inspector



TV node controls

Controls Tab

The Controls tab is the first of three tabs used to customize the analog TV distortion. The Controls tab modifies the scan lines and image distortion of the effect.

Scan Lines

This slider is used to emulate the interlaced look by dropping lines out of the image. Setting it to black, with a transparent Alpha, drops a line. A value of 1 (default) drops every second line. A value of 2 shows one line, and then drops the second and third and repeats. A value of zero turns off the effect.

Horizontal

Use this slider to apply a simple Horizontal offset to the image.

Vertical

Use this slider to apply a simple Vertical offset to the image.

Skew

This slider is used to apply a diagonal offset to the image. Positive values skew the image to the top left. Negative values skew the image to the top right. Pixels pushed off frame wrap around and reappear on the other side of the image.

Amplitude

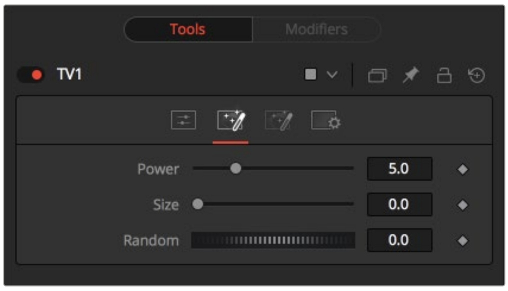
The Amplitude slider can be used to introduce smooth sine wave-type deformation to the edges of the image. Higher values increase the intensity of the deformation. Use the Frequency control to determine how often the distortion is repeated.

Frequency

The Frequency slider sets the frequency of the sine wave used to produce distortion along the edges of the image when the amplitude control is greater than 1.

Offset

Use Offset to adjust the position of the sine wave, causing the deformation applied to the image via the Amplitude and Frequency controls to see across the image.



The TV Noise tab

Noise Tab

The Noise tab is the second of three tabs used to customize the analog TV distortion. The Noise tab modifies the noise in the image to simulate a weak analog antenna signal.

Power

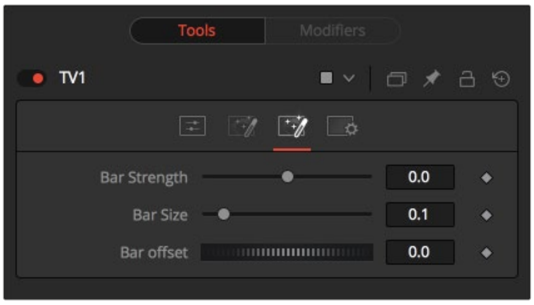
Increase the value of this slider above 0 to introduce noise into the image. The higher the value, the stronger the noise.

Size

Use this slider to scale the noise map larger.

Random

If this thumbwheel control is set to 0, the noise map is static. Change the value over time to cause the static to change from frame to frame.



The TV Roll Bar tab

Roll Bar Tab

The Roll Bar tab is the third of three tabs used to customize the analog TV distortion. The Roll Bar tab animates the bar.

Bar Strength

At the default value of 0, no bar is drawn. The higher the value, the darker the area covered by the bar becomes.

Bar Size

Increase the value of this slider to make the bar taller.

Bar Offset

Animate this control to scroll the bar across the screen.

Common Controls

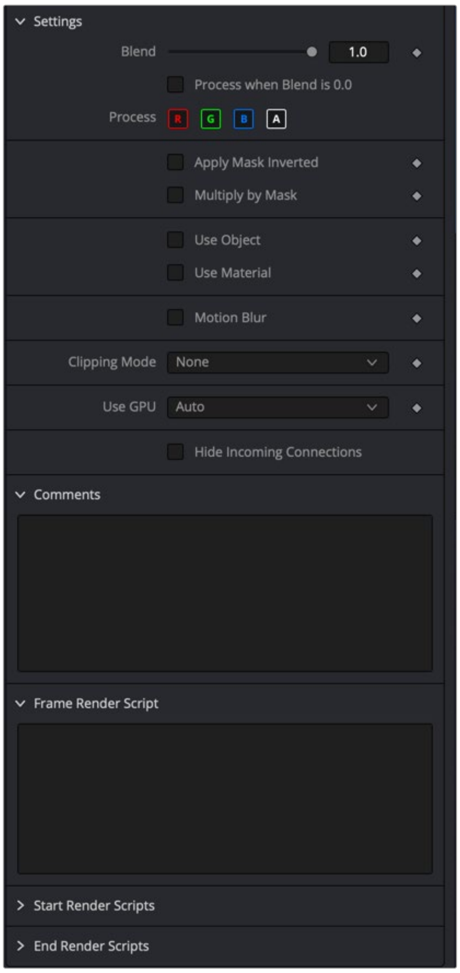
Settings Tab

The Settings tab controls are common to all Effect nodes, so their descriptions can be found in the following "The Common Controls" section.

The Common Controls

Effect nodes share several identical controls in the Inspector. This section describes controls that are common among Effect nodes.

Inspector



The Common Effects Settings tab

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Effects category. The Settings controls are even found on third-party Effects-type plugin tools. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options, which are also covered here.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. This causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This is useful when this node is scripted to trigger another task, but the blend is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this possess a set of like RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Control tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

> For more information on coverage and background channels, *see Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Clipping Mode

This option determines how the domain of definition rendering handles edges. The Clipping mode is most important when blur or softness is applied, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **None:** Setting this option to None does not perform any source image clipping. Any data required to process the node's effect that would usually be outside the upstream DoD is treated as black/transparent.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one full frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.